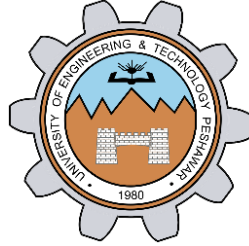


Investigating the TCP/IP and OSI Models in Action

Assignment # 02



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CSE-303 Data Communication & Networks

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Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Date:

13th May 2024

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Objectives

Part 1: Examine HTTP Web Traffic

Part 2: Display Elements of the TCP/IP Protocol Suite

Background

This simulation activity is intended to provide a foundation for understanding the TCP/IP protocol suite and the relationship to the OSI model. Simulation mode allows you to view the data contents being sent across the network at each layer.

As data moves through the network, it is broken down into smaller pieces and identified so that the pieces can be put back together when they arrive at the destination. Each piece is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer of the TCP/IP and OSI models. Packet Tracer simulation mode enables you to view each of the layers and the associated PDU. The following steps lead the user through the process of requesting a web page from a web server by using the web browser application available on a client PC.

Even though much of the information displayed will be discussed in more detail later, this is an opportunity to explore the functionality of Packet Tracer and be able to visualize the encapsulation process.

Instructions

Part 1: Examine HTTP Web Traffic

In Part 1 of this activity, you will use Packet Tracer (PT) Simulation mode to generate web traffic and examine HTTP.

Step 1: Switch from Realtime to Simulation mode.

In the lower right corner of the Packet Tracer interface are buttons that toggle between **Realtime** and **Simulation** mode. PT always starts in **Realtime** mode, in which networking protocols operate with realistic timings. However, a powerful feature of Packet Tracer allows the user to “stop time” by switching to Simulation mode. In Simulation mode, packets are displayed as animated envelopes, time is event driven, and the user can step through networking events.

- a. Click the **Simulation** mode icon to switch from **Realtime** mode to **Simulation** mode.

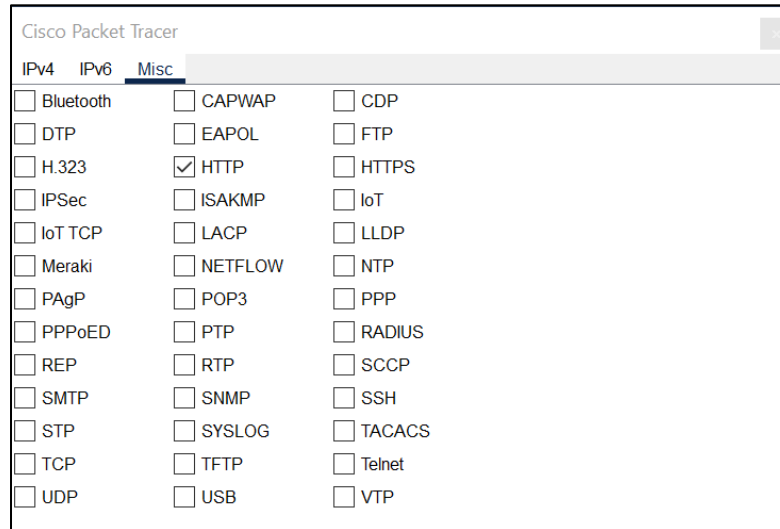


- b. Select **HTTP** from the **Event List Filters**.

- 1) HTTP may already be the only visible event. If necessary, click the **Edit Filters** button at the bottom of the simulation panel to display the available visible events. Toggle the **Show**

All/None check box and notice how the check boxes switch from unchecked to checked or checked to unchecked, depending on the current state.

- 2) Click the **Show All/None** check box until all boxes are cleared and then select **HTTP** from the Misc tab of the Edit Filters window. Click the X in the upper right hand corner of the window to close the **Edit Filters** window. The Visible Events should now only display HTTP.

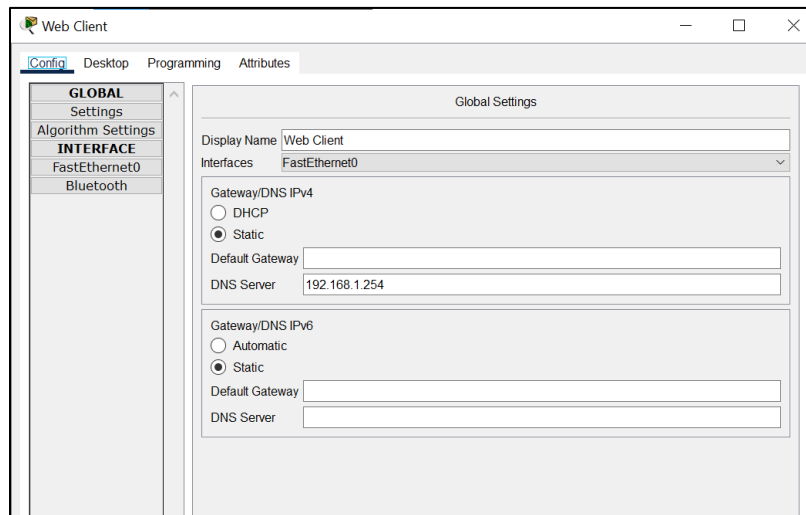


Step 2: Generate web (HTTP) traffic.

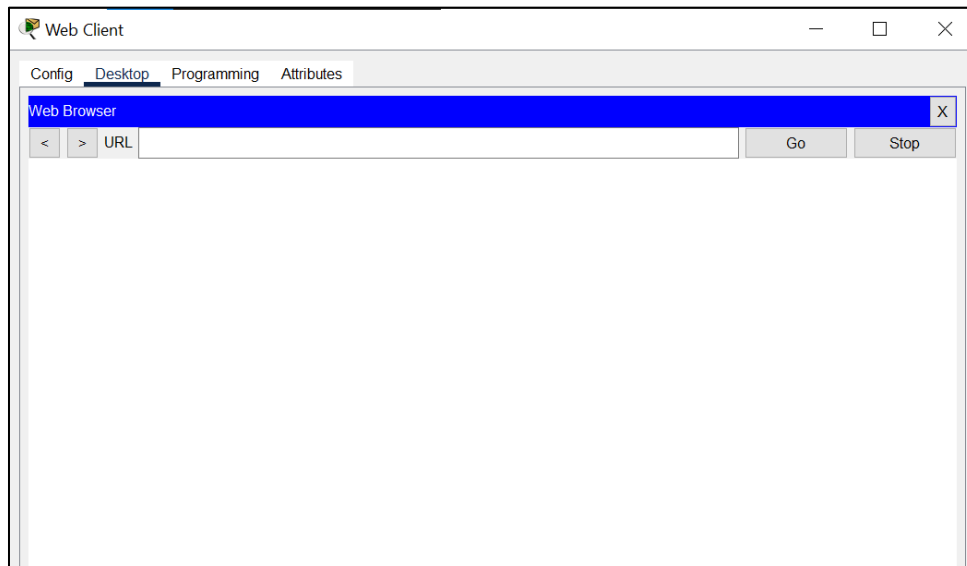
Currently the Simulation Panel is empty. There are five columns listed across the top of the Event List within the Simulation Panel. As traffic is generated and stepped through, events appear in the list.

Note: The Web Server and Web Client are displayed in the left pane. The panels can be adjusted in size by hovering next to the scroll bar and dragging left or right when the double-headed arrow appears.

- a. Click **Web Client** in the far left pane.

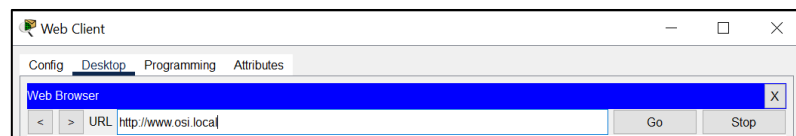


- b. Click the **Desktop** tab and click the **Web Browser** icon to open it.



- c. In the URL field, enter **www.osi.local** and click **Go**.

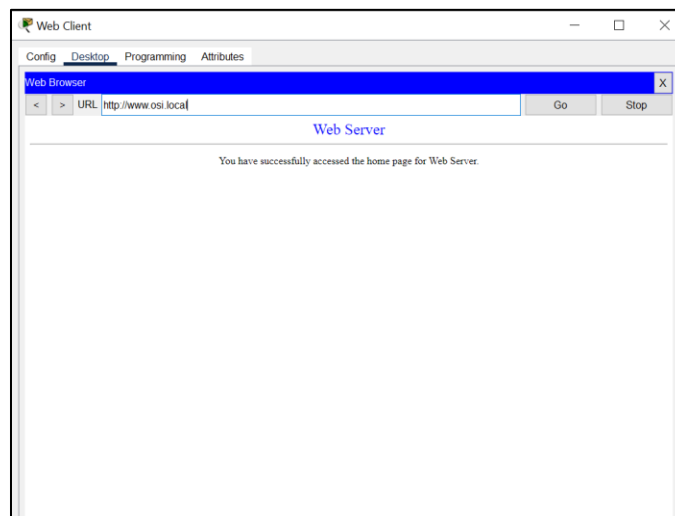
Because time in Simulation mode is event-driven, you must use the **Capture/Forward** button to display network events. The capture forward button is located at the left hand side of the blue band that is below the topology window. Of the three buttons there, it is the one on the right.



- d. Click **Capture/Forward** four times. There should be four events in the Event List.

Look at the Web Client web browser page. Did anything change?

Ans: Yes, the requested web page was received (from the Web Server) at the Web Client.



Simulation Panel				
Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.007	--	Web Client	HTTP
	0.008	--	Web Client	HTTP
	0.009	Web Client	Web Server	HTTP
	0.010	Web Server	Web Client	HTTP

- e. Click the first colored square box under the **Event List > Type** column. It may be necessary to expand the **Simulation Panel** or use the scrollbar directly below the **Event List**.

The **PDU Information at Device: Web Client** window displays. In this window, there are only two tabs (**OSI Model** and **Outbound PDU Details**) because this is the start of the transmission. As more events are examined, there will be three tabs displayed, adding a tab for **Inbound PDU Details**. When an event is the last event in the stream of traffic, only the **OSI Model** and **Inbound PDU Details** tabs are displayed.

PDU Information at Device: Web Client

OSI Model
Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: HTTP CLIENT

In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1

Out Layers
Layer 7: HTTP
Layer6
Layer5
Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer 1: Port(s):

1. The HTTP client sends a HTTP request to the server.

Challenge Me
<< Previous Layer
Next Layer >>

- f. Ensure that the **OSI Model** tab is selected.
- Under the **Out Layers** column, click **Layer 7**.

What information is listed in the numbered steps directly below the **In Layers** and **Out Layers** boxes for Layer 7?

Ans: It is giving us the information about the protocol used in this layer. In this case, the Web Client is sending an HTTP request to the Web Server. To send a successful HTTP request, HTTP protocol is used in Layer 7 (Application Layer).

What is the **Dst Port** value for **Layer 4** under the **Out Layers** column?

Ans:

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: HTTP CLIENT

In Layers	Out Layers
Layer7	Layer 7: HTTP
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer3	Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer2	Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer1	Layer 1: Port(s):

1. Sent segment information: the sequence number 1, the ACK number 1, and the data length 102.

Challenge Me << Previous Layer Next Layer >>

What is the **Dest. IP** value for **Layer 3** under the **Out Layers** column?

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: HTTP CLIENT

In Layers	Out Layers
Layer7	Layer 7: HTTP
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer3	Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer2	Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer1	Layer 1: Port(s):

1. The destination IP address is in the same subnet. The device sets the next-hop to destination.

Challenge Me << Previous Layer Next Layer >>

What information is displayed at Layer 2 under the **Out Layers** column?

Ans: Layer 2 Ethernet II Header, inbound and outbound MAC addresses are displayed at Layer 2.

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: HTTP CLIENT

In Layers	Out Layers
Layer7	Layer 7: HTTP
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer3	Layer 3: IP Header Src. IP: 192.168.1.1, Dst. IP: 192.168.1.254
Layer2	Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer1	Layer 1: Port(s):

1. The next-hop IP address is a unicast. The ARP process looks it up in the ARP table.
2. The next-hop IP address is in the ARP table. The ARP process sets the frame's destination MAC address to the one found in the table.
3. The device encapsulates the PDU into an Ethernet frame.

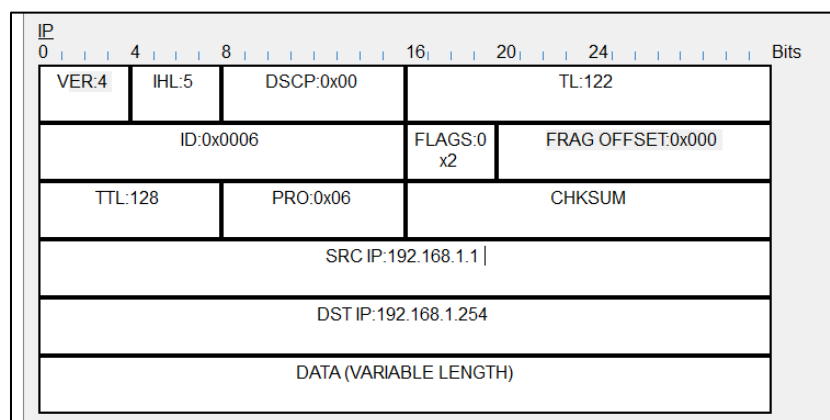
Challenge Me << Previous Layer Next Layer >>

- g. Click the **Outbound PDU Details** tab.

Information listed under the **PDU Formats** is reflective of the layers within the TCP/IP model.

Note: The information listed under the **Ethernet II** section of the Outbound PDU Details tab provides even more detailed information than is listed under Layer 2 on the **OSI Model** tab. The **Outbound PDU Details** provides more descriptive and detailed information. The values under **DEST MAC** and **SRC MAC** within the **Ethernet II** section of the **PDU Details** appear on the **OSI Model** tab under Layer 2, but are not identified as such. Questions:

What is the common information listed under the **IP** section of **PDU Details** as compared to the information listed under the **OSI Model** tab? With which layer is it associated?



Ans: IP section is associated with the Network Layer (Layer 5 of OSI Model). Source IP and Destination IP are commonly listed under both PDU details and OSI Model tab. However, the PDU details section contains a lot of information. For example, VERSION, IHL(Internet Header

Length), DSCP(Differentiated Services Code Point), TL(Total Length), ID, FLAGS, FRAG OFFSET, TTL(Time to Live), PRO(Protocol Parameter), Checksum, SRC IP, DST IP and DATA.

What is the common information listed under the **TCP** section of **PDU Details**, as compared to the information listed under the **OSI Model** tab, and with which layer is it associated?

Ans: **TCP** section is associated with the Transport Layer (Layer 4 of OSI Model). Source Port and Destination Port are commonly listed under both PDU details and OSI Model tab. The PDU details section of TCP contains a lot of information like Sequence Number, Acknowledgement Number, OFFSET etc.

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: HTTP CLIENT

In Layers	Out Layers
Layer7	Layer 7: HTTP
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer3	Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer2	Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer1	Layer 1: Port(s):

1. The destination IP address is in the same subnet. The device sets the next-hop to destination.

Challenge Me << Previous Layer Next Layer >>

TCP

0 4 8 16 24 Bits

SOURCE PORT:1025		DESTINATION PORT:80	
SEQUENCE NUMBER:1			
ACKNOWLEDGEMENT NUMBER:1			
OFFSE T:0x0	RESER VED: 0	FLAGS:0b000110 00	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

What is the **Host** listed under the **HTTP** section of the **PDU Details**? What layer would this information be associated with under the **OSI Model** tab?

HTTP REQUEST

0 4 8 16 Bytes

Connection: close

Host: www.osi.local

Ans: Host is www.osi.local

This information would be associated with layer 7 (Application Layer) under the OSI Model Tab.

- h. Click the next colored square box under the **Event List > Type** column. Only Layer 1 is active (not grayed out). The device is moving the frame from the buffer and placing it on to the network.

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: HTTP CLIENT

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer3
Layer2	Layer2
Layer1	Layer 1: Port(s): FastEthernet0

1. The device takes out this frame from the buffer and sends it.
2. FastEthernet0 sends out the frame.

Challenge Me << Previous Layer Next Layer >>

- i. Advance to the next HTTP **Type** box within the **Event List** and click the colored square box. This window contains both **In Layers** and **Out Layers**. Notice the direction of the arrow directly under the **In Layers** column; it is pointing upward, indicating the direction the data is travelling. Scroll through these layers making note of the items previously viewed. At the top of the column the arrow points to the right. This denotes that the server is now sending the information back to the client.

Comparing the information displayed in the **In Layers** column with that of the **Out Layers** column, what are the major differences?

Ans:

Layer 4 (Transport Layer):

- **In Layers:** The TCP source port is 1025 while destination port is 80.
- **Out Layers:** The TCP source port is 80 while destination port is 1025.

Layer 3 (Network Layer):

- **In Layers:** Source IP address is 192.168.1.1 and Destination IP address is 192.168.1.254

- **Out Layers:** Source IP address is 192.168.1.254 and Destination IP address is 192.168.1.1

Layer 2 (Data Link Layer - Ethernet II Header):

- **In Layers:** The Ethernet II Header addresses are 0060.47CA.4DEE (source) and 0001.9649.401D (destination).
- **Out Layers:** The Ethernet II Header addresses are reversed: 0001.9649.401D (source) and 0060.47CA.4DEE (destination).

Other layers (Layer 7, 6, 5 and 1) appear to be consistent between the “In Layers” and “Out Layers” columns

PDU Information at Device: Web Server

OSI Model
Inbound PDU Details
Outbound PDU Details

At Device: Web Server
Source: Web Client
Destination: HTTP CLIENT

In Layers	Out Layers
Layer 7: HTTP	Layer 7: HTTP
Layer6	Layer6
Layer5	Layer5
Layer 4: TCP Src Port: 1025, Dst Port: 80	Layer 4: TCP Src Port: 80, Dst Port: 1025
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254	Layer 3: IP Header Src. IP: 192.168.1.254, Dest. IP: 192.168.1.1
Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D	Layer 2: Ethernet II Header 0001.96A9.401D >> 0060.47CA.4DEE
Layer 1: Port FastEthernet0	Layer 1: Port(s): FastEthernet0

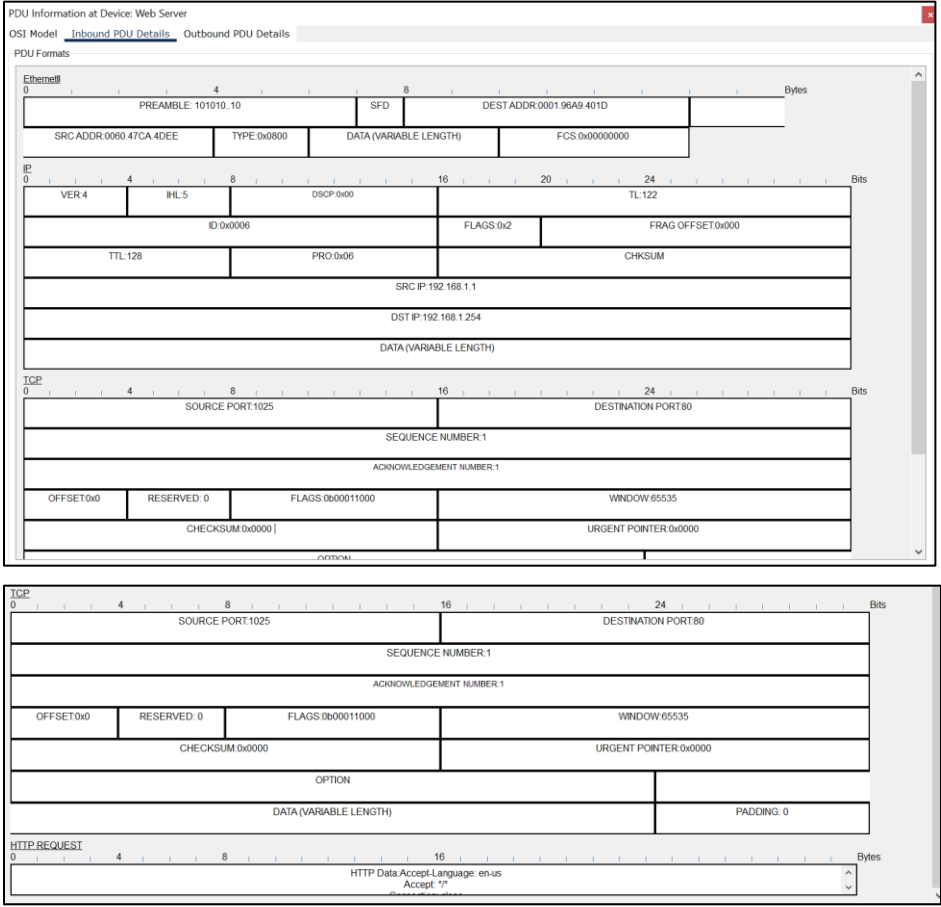
1. FastEthernet0 receives the frame.

Challenge Me
<< Previous Layer
Next Layer >>

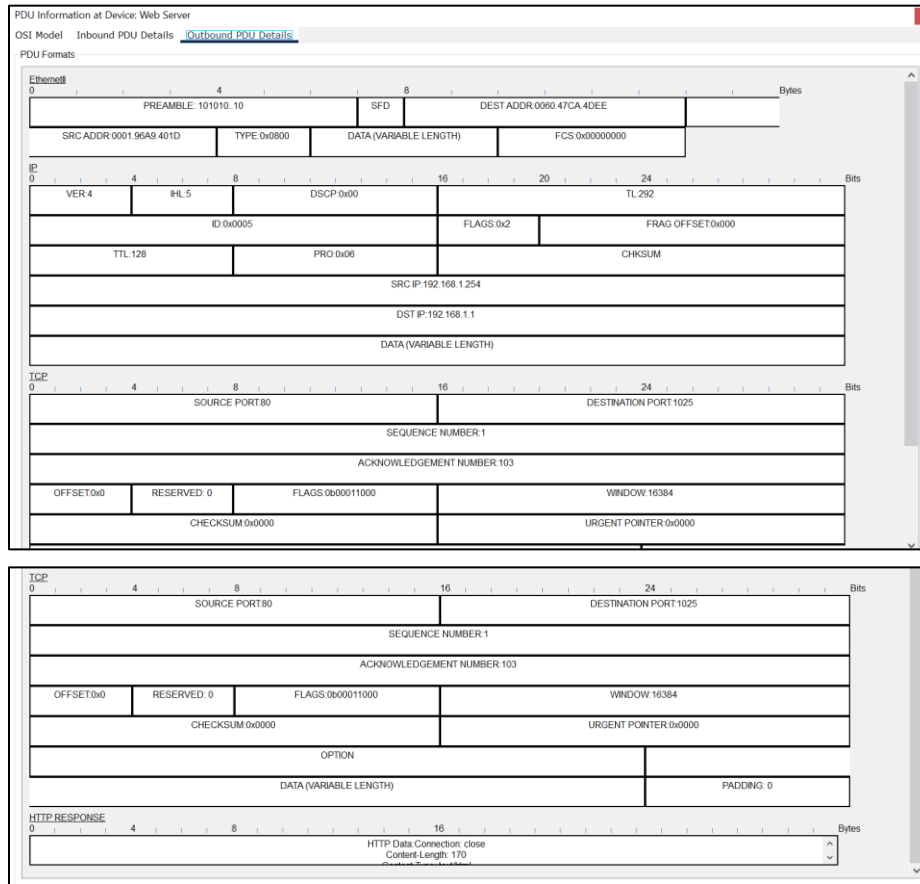
j. Click the **Inbound and Outbound PDU Details** tab. Review the PDU details.

Ans: The Inbound and Outbound PDU details tab shows the process of De-Encapsulation and Encapsulation (in this case). We can see a lot of information and various headers added by each layer.

Inbound PDU Details:



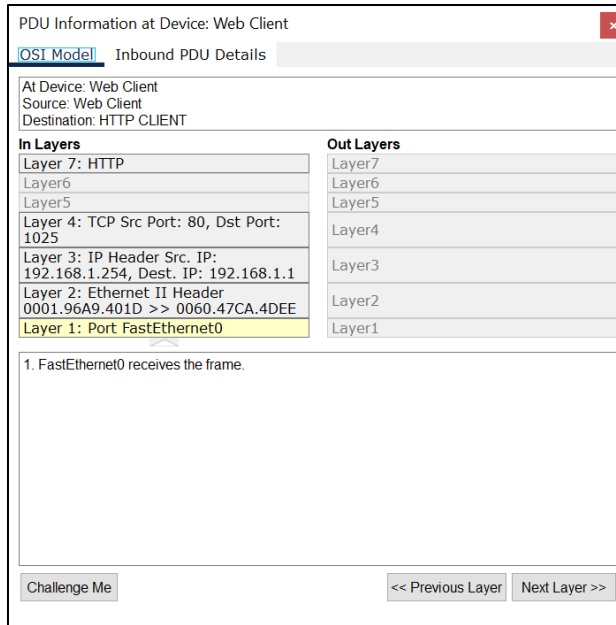
Outbound PDU Details:



k. Click the last colored square box under the **Info** column.

How many tabs are displayed with this event? Explain.

Ans: Two tabs **OSI Model** and **Inbound PDU Details** were displayed with this event.



Part 2: Display Elements of the TCP/IP Protocol Suite

In Part 2 of this activity, you will use the Packet Tracer Simulation mode to view and examine some of the other protocols comprising of TCP/IP suite.

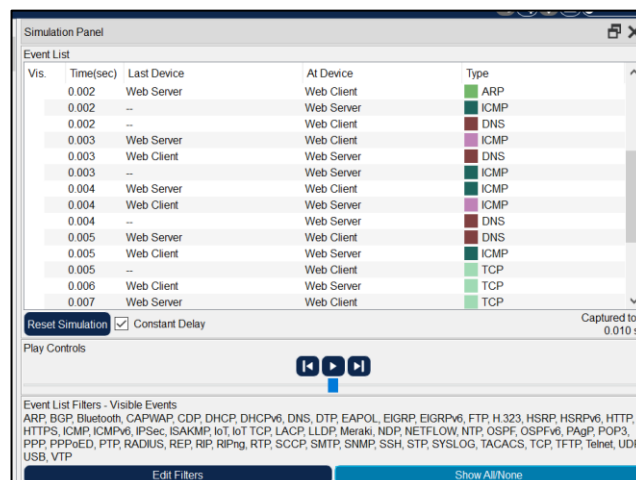
Step 1: View Additional Events

- Close any open PDU information windows.
- In the **Event List Filters > Visible Events** section, click **Show All/None**.



What additional Event Types are displayed?

Ans: The Additional Event Types are **ARP, DNS, ICMP and TCP**.



These extra entries play various roles within the TCP/IP suite. Address Resolution Protocol (ARP) requests MAC addresses for destination hosts. DNS is responsible for converting a name (for example, **www.osi.local**) to an IP address. The additional TCP events are responsible for connecting, agreeing on communication parameters, and disconnecting the communications sessions between the devices. These protocols have been mentioned previously and will be further discussed as the course progresses. Currently there are over 35 possible protocols (event types) available for capture within Packet Tracer.

- c. Click the first DNS event in the **Type** column. Explore the **OSI Model** and **PDU Detail** tabs and note the encapsulation process. As you look at the **OSI Model** tab with **Layer 7** highlighted, a description of what is occurring is listed directly below the **In Layers** and **Out Layers** ("1. The DNS client sends a DNS query to the DNS server."). This is very useful information to help understand what is occurring during the communication process.

PDU Information at Device: Web Client

OSI Model
Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: 192.168.1.254

In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1

Out Layers
Layer 7: DNS
Layer6
Layer5
Layer 4: UDP Src Port: 1025, Dst Port: 53
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer 2:
Layer1

1. The DNS client sends an A DNS query to the DNS server.

Challenge Me
<< Previous Layer
Next Layer >>

- d. Click the **Outbound PDU Details** tab.

What information is listed in the **NAME** field: in the DNS QUERY section?

Ans: www.osi.local

PDU Information at Device: Web Client

OSI Model
Outbound PDU Details

PDU Formats

DATA (VARIABLE LENGTH)

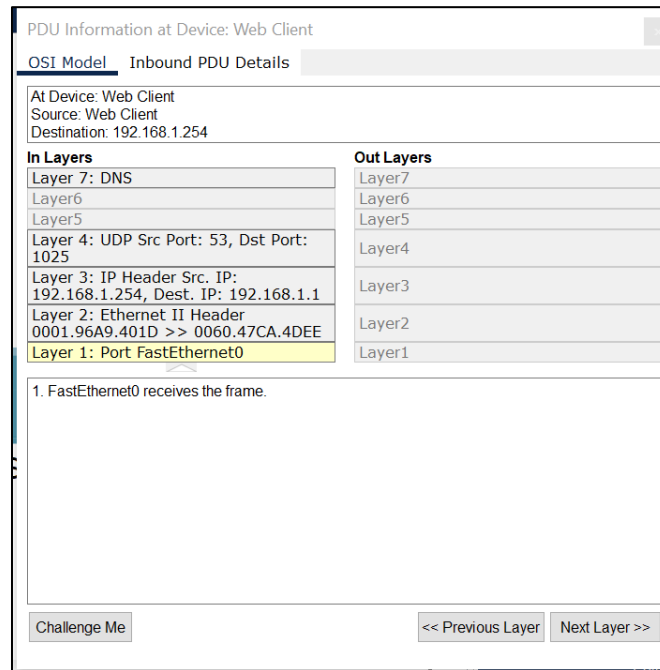
DNS Header
Transaction ID 0xc099
OPCODE 0x1
Z
RCODE 0x0
QDCOUNT 1
ANCOUNT 0
NSCOUNT 0
ARCOUNT 0

DNS Query
NAME (VARIABLE LENGTH) www.osi.local
TYPE 1
CLASS 1
TTL 66400
LENGTH 0

- e. Click the last DNS **Info** colored square box in the event list.

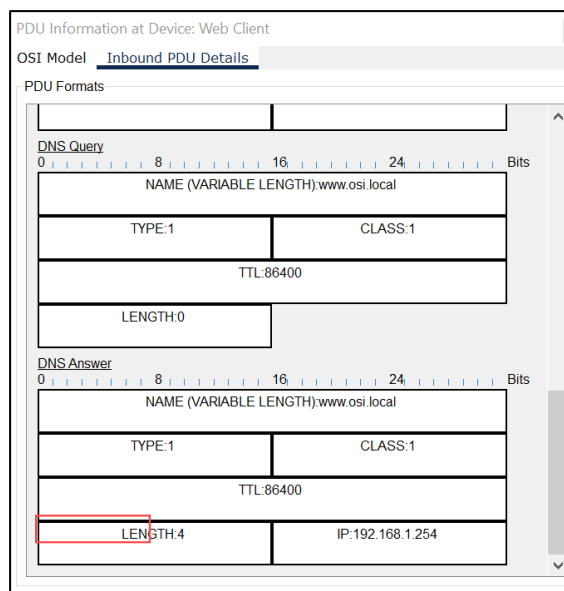
At which device was the PDU captured?

Ans: The Web Client



What is the value listed next to **ADDRESS:** in the DNS ANSWER section of the **Inbound PDU Details**?

Ans: 192.168.1.254 which is the address of Web Server.



- f. Find the first **HTTP** event in the list and click the colored square box of the **TCP** event immediately following this event. Highlight **Layer 4** in the **OSI Model** tab.

In the numbered list directly below the **In Layers** and **Out Layers**, what is the information displayed under items 4 and 5?

Ans:

4. The TCP connection is successful.

5. The device sets the connection state to ESTABLISHED.

PDU Information at Device: Web Server

OSI Model Inbound PDU Details

At Device: Web Server
Source: Web Client
Destination: 192.168.1.254

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer 4: TCP Src Port: 1025, Dst Port: 80	Layer4
Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254	Layer3
Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D	Layer2
Layer 1: Port FastEthernet0	Layer1

1. The device receives a TCP ACK segment on the connection to 192.168.1.1 on port 1025.
2. Received segment information: the sequence number 1, the ACK number 1, and the data length 20.
3. The TCP segment has the expected peer sequence number.
4. The TCP connection is successful.
5. The device sets the connection state to ESTABLISHED.

Challenge Me << Previous Layer Next Layer >>

TCP manages the connecting and disconnecting of the communications channel along with other responsibilities. This particular event shows that the communication channel has been ESTABLISHED.

- g. Click the last TCP event. Highlight Layer 4 in the **OSI Model** tab. Examine the steps listed directly below **In Layers** and **Out Layers**.

What is the purpose of this event, based on the information provided in the last item in the list (should be item 4)?

Ans: Its purpose is closing the connection.

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: 192.168.1.254

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer3	Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer2	Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer1	Layer 1: Port(s): FastEthernet0

1. The device closes the TCP connection to 192.168.1.254 on port 80.
2. The device sets the connection state to FIN_WAIT_1.
3. The device sends a TCP FIN+ACK segment.
4. Sent segment information: the sequence number 103, the ACK number 273, and the data length 20.

Challenge Me << Previous Layer Next Layer >>

Challenge Questions

This simulation provided an example of a web session between a client and a server on a local area network (LAN). The client makes requests to specific services running on the server. The server must be set up to listen on specific ports for a client request. (Hint: Look at Layer 4 in the **OSI Model** tab for port information.)

Based on the information that was inspected during the Packet Tracer capture, what port number is the **Web Server** listening on for the web request?

Ans: Based on the information that was inspected during the Packet Tracer capture, **Web Server** was listening on for the web request on **Port 80**.

PDU Information at Device: Web Client

OSI Model Outbound PDU Details

At Device: Web Client
Source: Web Client
Destination: HTTP CLIENT

In Layers	Out Layers
Layer7	Layer 7: HTTP
Layer6	Layer6
Layer5	Layer5
Layer4	Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer3	Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254
Layer2	Layer 2: Ethernet II Header 0060.47CA.4DEE >> 0001.96A9.401D
Layer1	Layer 1: Port(s):

1. The HTTP client sends a HTTP request to the server.

What port is the **Web Server** listening on for a DNS request?

Ans: The **Web Server** is listening on **port 53** for DNS request.

PDU Information at Device: Web Client

OSI Model

Outbound PDU Details

At Device: Web Client

Source: Web Client

Destination: 192.168.1.254

In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer1

Out Layers

Layer 7: DNS

Layer6

Layer5

Layer 4: UDP Src Port: 1025, Dst Port: 53

Layer 3: IP Header Src. IP: 192.168.1.1, Dest. IP: 192.168.1.254

Layer 2:

Layer1

1. The device encapsulates the PDU into an UDP segment.